

PRODUCT STRATEGY · Q3 2025

From Signal to Strategy

A decision framework for product leaders navigating market uncertainty

SECTION 01 · FOUNDATIONS

The landscape has shifted. Here is
what that means for us.

MODULE 02

Before we score signals, we need to agree on what a signal is.

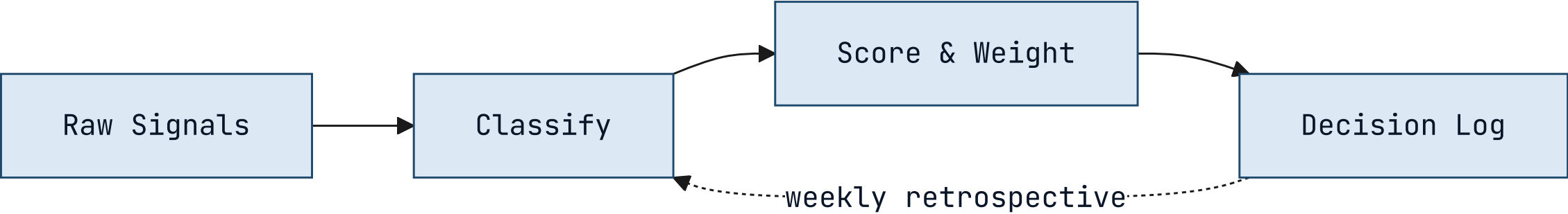
The word is overloaded. We use it to mean anything from a customer complaint to a macro trend. This framework requires a tighter definition.

The window for differentiation is narrowing.

Three converging forces — commoditized infrastructure, compressed release cycles, and rising customer switching costs — have reduced the average durable advantage window from 36 months to under 14. Teams that cannot identify signal from noise in that window will consistently miss timing.

How signals move from input to decision.

Four-stage processing pipeline — weekly cadence



IMPACT · PILOT RESULTS

Six months of results across four product teams.

Measured against pre-framework baseline, same teams, same market conditions.

73%
FASTER CLOSE

4.2×
SIGNAL RECALL

18
DECISIONS LOGGED

91%
TEAM ALIGNMENT

The framework has four components.

Signal Intake

Weekly structured collection across customer conversations, market data, and competitive moves. Normalized into a common schema before scoring.

Scoring Model

Each signal scored on three dimensions: confidence, recency, and strategic relevance. Weights are team-configurable and reviewed quarterly.

Decision Log

Every decision recorded with the signals that informed it, the options considered, and the criteria applied. Feeds the calibration loop.

Calibration Loop

Monthly retrospective that compares predicted outcomes to actual outcomes and adjusts scoring weights accordingly.

Code in card headers and body text.

Signal Intake

v2.4

Handles 94% of structured signals without manual intervention. Average latency: 4 min from ingestion to scored entry.

Decision Log

required

Every prioritization change above P2 must carry a logged rationale. No log, no change.

Scoring Model

configurable

Three dimensions: confidence, recency, relevance. Default weights are 33 / 33 / 33 — adjust after your first retrospective.

Calibration Loop

monthly

Compares predicted outcomes to actuals. First meaningful weight update happens after 2 cycles.

Signal Intake produces three outputs.

1

Weekly Signal Brief

A ranked list of the top 10 signals from the prior week, with confidence scores and source attribution. Distributed to product leads every Monday morning.

2

Anomaly Alerts

Real-time flags when a signal exceeds the 2σ threshold on any dimension. Routed directly to the accountable PM with a 4-hour response SLA.

3

Monthly Signal Index

The source of truth for the calibration loop. A complete record of all signals logged, scored, and resolved in the prior month. Required reading before each retrospective.

Two failure modes the framework is designed to prevent.

False signal amplification. A single loud voice — one enterprise customer, one analyst report, one competitive announcement — dominates the decision without being weighed against the full signal set. The scoring model prevents any single source from exceeding 30% of the total signal weight in a given decision.

Signal hoarding. Teams collect signals but do not log decisions, so the calibration loop has nothing to learn from. The Decision Log is a required artifact for any prioritization change above P2 severity. No log, no change.

Two intake modes for different signal types.

Structured Intake

Signals with clear schema: NPS verbatims, support ticket categories, feature request volumes, win/loss notes. Ingested automatically via API connectors. Scored on arrival. Zero manual handling.

Unstructured Intake

Signals without schema: field observations, conference conversations, analyst briefings, competitive demos. Require human classification before scoring. Routed to the signal owner for a 48-hour classification window.

Scoring model: before and after the calibration loop.

Before Calibration

Equal weights across all three dimensions. Confidence, recency, and relevance each contribute 33% to the final score. Simple, consistent, but blind to what your market actually rewards.



After Calibration

Weights reflect your team's historical signal accuracy. If recency has consistently been the weakest predictor for your product, it gets downweighted. The model becomes a record of what you have learned.

The shift from equal weights to calibrated weights takes two retrospective cycles — roughly 60 days from adoption.

“

The signal was always there. We just didn't have a system that forced us to look at it before we'd already decided.

”

— Head of Product, Pilot Team 3

How a decision moves through the framework.



What the framework does not do.

- It does not make decisions — it structures the information that humans use to decide.
- It does not replace customer discovery — it scores and routes what discovery surfaces.
- It does not work without the Decision Log — calibration requires outcome data to learn from.
- It does not guarantee alignment — it surfaces disagreement earlier, which still requires resolution.
- It does not scale down to individual feature decisions — it is designed for prioritization above P2.

Four things that must be true before you begin.

- 01 You have a regular prioritization cadence — at minimum monthly.
- 02 At least one person owns signal collection full-time or as a primary responsibility.
- 03 Leadership has agreed to log decisions with rationale, not just outcomes.
- 04 You have 90 minutes per week to run the intake and scoring process.

CALIBRATION RESULT • 6-MONTH PILOT

14X

Return on signal investment — measured as decisions that reached the right outcome on the first attempt, versus the baseline rate before the framework was adopted.



SECTION 02

Scoring Model Deep Dive

Dark panel + content · split-panel

WHAT THIS SECTION COVERS

The scoring model is the most configurable component. This section covers the three dimensions, how weights are set initially, and how calibration updates them over time.

1

Confidence

How many independent sources corroborate the signal. Ranges 1–5.

2

Recency

Time-decay applied from signal date to scoring date. Half-life is team-configurable.

3

Strategic Relevance

Manual score from the signal owner. Ranges 1–5. Requires justification above 4.

WHAT WOULD HELP US MOVE FORWARD

Next step is a working session, not a debate.

*Walk these questions with me in 60–90 minutes. The output is either
a design we can execute, or a shared list of what needs more work
before we commit.*

FINDING 01 · STRUCTURED INTAKE

Structured intake performed above expectations — volume and latency were not concerns.

What worked

API connectors handled 94% of structured signals without manual intervention. Average scoring latency was 4 minutes from ingestion. Schema normalization held across all five connected sources.

What required tuning

NPS verbatim classification had an 18% error rate in the first two weeks. Required a training pass on the classification model before accuracy reached the 92% target.

KEY INSIGHT

Viable as designed — NLP classification requires a 2-week warm-up period on new deployments.

Key insight works on any card-bearing layout.

Signal Intake

Weekly structured collection across customer conversations, market data, and competitive moves.

Scoring Model

Each signal scored on three dimensions: confidence, recency, and strategic relevance.

Decision Log

Every decision recorded with the signals that informed it and the criteria applied.

Calibration Loop

Monthly retrospective that compares predicted outcomes to actual outcomes.

KEY INSIGHT

The calibration loop is what separates teams that learn from teams that repeat the same mistakes.

¹ Trailing blockquote becomes a key insight. Trailing paragraph becomes a below-note.

Three scoring failure modes found in the pilot.

1 Recency dominance

High-recency noise crowding out durable signal. Teams set recency weight above 50% in the first calibration pass. Corrected by capping recency weight at 40% until two calibration cycles complete.

2 Source concentration

Single-customer signals inflating confidence scores. One enterprise customer's verbatims represented 34% of all structured intake in month one. Corrected by adding a source-diversity floor to the scoring model.

3 Outcome misclassification

PMs logging predicted outcomes that were too vague to score at retrospective. "Improve retention" is not scoreable. "Reduce 30-day churn from 8.2% to below 7%" is.

Four requirements every decision system must meet.

01

Speed

Decisions must close within the window they are relevant to. Systems that add latency consume the value they exist to protect.

02

Auditability

Every prioritization decision above a threshold must carry a traceable rationale. Required for alignment and compliance.

03

Adoption

If the team won't use it weekly, calibration never runs and the model never improves. Ninety minutes per PM is the ceiling.

04

Calibration

The system must improve over time. A static scoring model is a spreadsheet with extra steps.

We evaluated four intake tools against the criteria.

Tool A · Chorus

- ✓ Speed
- ~ Auditability
- ✓ Adoption
- ✗ Calibration

Strong call recording and summarization. No decision logging or calibration loop. Requires separate tooling for everything downstream of intake.

Tool B · Productboard

- ✗ Speed
- ✓ Auditability
- ✓ Adoption
- ✗ Calibration

Solid intake and prioritization. Decision logging exists but is manual and rarely used. No calibration mechanism. Setup takes 3–4 weeks.

Tool C · Notion

- ✓ Speed
- ✓ Auditability
- ~ Adoption
- ✗ Calibration

Flexible enough to build the full system. But building it takes 40+ hours and the result is fragile. Teams abandon maintenance after the first quarter.

Tool D · Sprig + Decision Log



- ✓ Speed
- ✓ Auditability
- ✓ Adoption
- ✓ Calibration

Meets all four criteria within the 90-minute weekly budget. Reaches production in the same week it is adopted. Recommended.

The four tools side by side.

CRITERION	CHORUS	PRODUCTBOARD	NOTION	SPRIG + LOG
Speed	✓	✗	✓	✓
Auditability	✗	✓	✓	✓
Adoption	✓	✓	✗	✓
Calibration	✗	✗	✗	✓
Setup time	1 day	3–4 weeks	40+ hours	Same day

Evaluated against the same four teams and the same 90-minute weekly budget constraint.

Applying the criteria to the tools — here is where the evidence points.

THE EVIDENCE FAVORS TOOL D



Sprig combined with a lightweight Decision Log meets all four criteria within the 90-minute weekly budget, reaches production in the same week it is adopted, and leaves a clean exit ramp if a better native solution emerges.

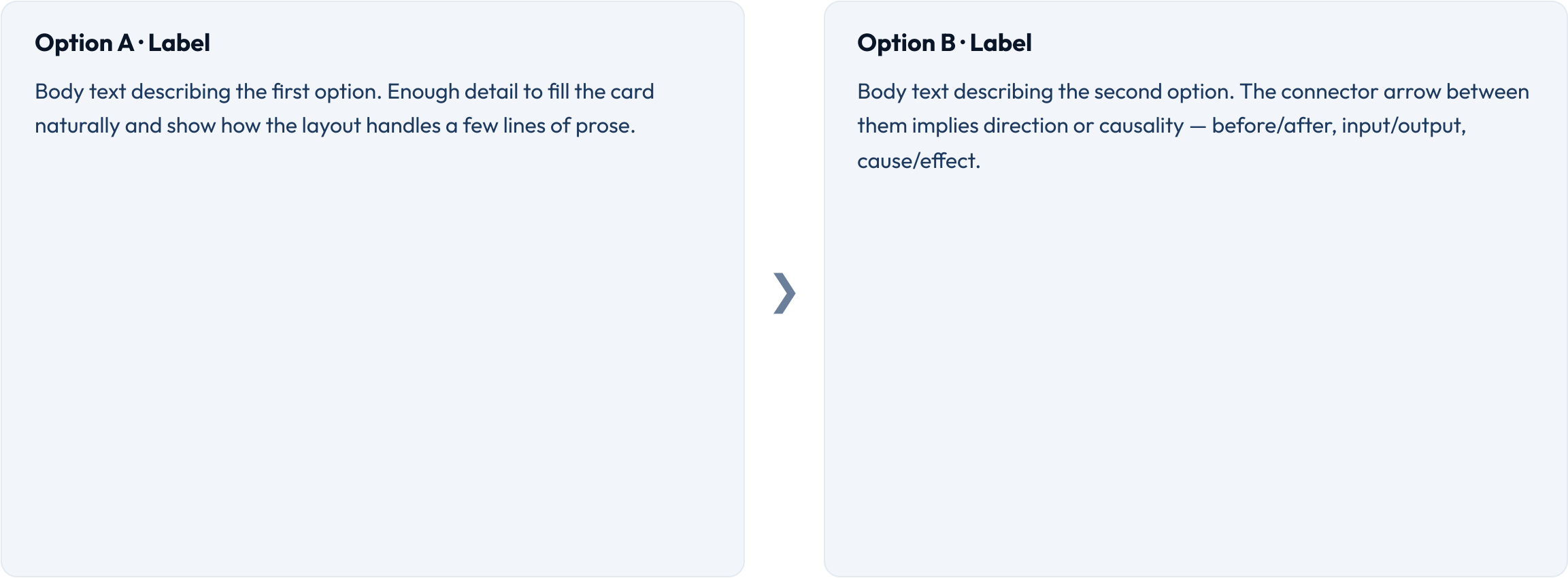
The path is not self-executing

Sprig requires a connector built to your NPS and support platforms. Budget 4–6 hours of engineering time in week one. After that, zero maintenance overhead.

The Decision Log is the hardest part

Not technically. Culturally. PMs need to log decisions with predicted outcomes before they close, not after. This is a habit change, not a tool change.

Two options with a connector and an explanatory note below.



How to roll this out across your organization.

STEP 01

Pick one team and one decision type

Start with a team that already has a regular prioritization rhythm. Apply the framework only to a single decision category for the first 30 days.



STEP 02

Log everything, decide nothing differently

In the first month, do not change how you make decisions. Just log signals and decisions as you would have made them anyway.



STEP 03

Run your first retrospective

At day 30, score the logged decisions against outcomes. This is where the model gets its first calibration pass.



STEP 04

Expand to a second team

With one retrospective complete, you have evidence. Use it to onboard the second team with real data, not promises.

The six signal dimensions, what they measure, and how they are scored.

01	Confidence	Number of independent sources corroborating the signal	1-5 · Auto-scored
02	Recency	Time-decay from signal date, configurable half-life	0.0-1.0 · Auto-scored
03	Relevance	Alignment to current strategic bets, owner-scored	1-5 · Manual
04	Reach	Number of customers or segments affected	1-5 · Auto-scored
05	Effort	Engineering and design cost to act on the signal	1-5 · Manual
06	Confidence delta	Change in confidence score since last scoring cycle	-5 to +5 · Auto

HEADER AND FOOTER

Header stays uppercase — footer renders as written.

Set `header:` and `footer:` in frontmatter for deck-level labels, or use per-slide comment directives. The header uses uppercase text-transform automatically, so you write it in any case. The footer renders exactly as written.

The tokenization call is three lines of application code.

```
import { TokenVault } from "@company/token-sdk";

const vault = new TokenVault({ keyFile: "./vault.key" });

// Tokenize at ingestion
const token = await vault.tokenize(ssn, { field: "ssn", tenant: "acme" });

// Detokenize only at point of use – every call is logged
const plaintext = await vault.detokenize(token, { requestor: "claims-svc" });
```

BEFORE & AFTER · KEY DISTRIBUTION

File-distributed keys versus vault-integrated keys.

BEFORE · FILE-DISTRIBUTED

```
# Key material on disk – anyone with  
# filesystem access can read it  
with open('./vault.key', 'rb') as f:  
    key = f.read()  
  
cipher = AES(key)  
token = cipher.encrypt(ssn)
```

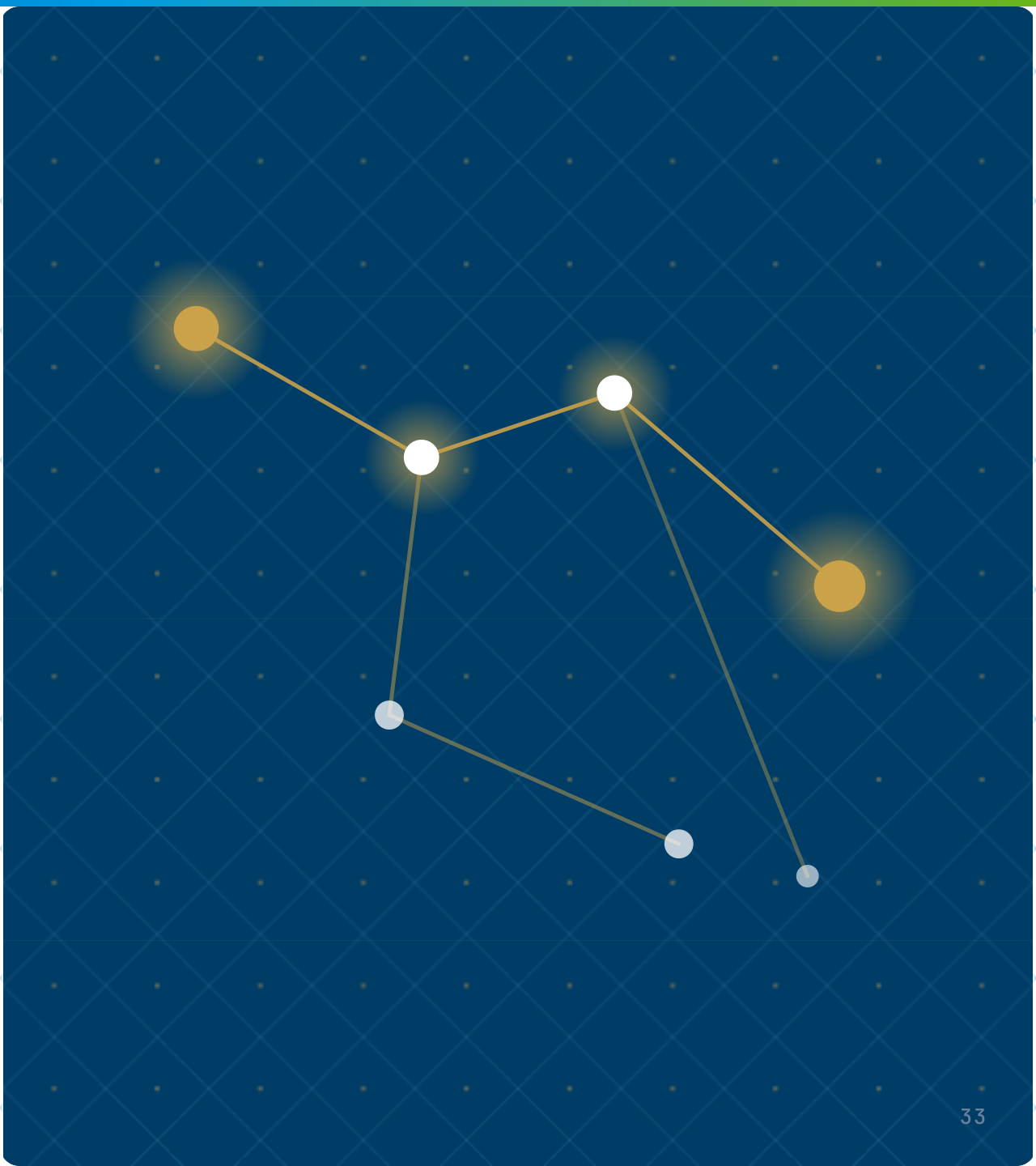
AFTER · HSM / KMS INTEGRATED

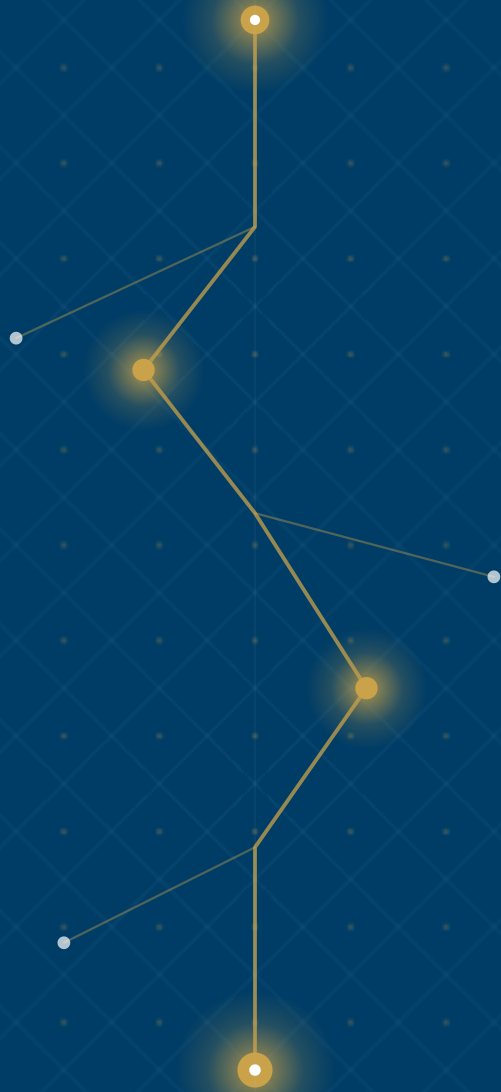
```
# Key never leaves the HSM –  
# every operation is audited  
import boto3  
  
kms = boto3.client('kms')  
token = kms.encrypt(  
    KeyId='alias/tokenization',  
    Plaintext=ssn  
)['CiphertextBlob']
```

LAYOUT • IMAGE

Image right is the default — text leads, evidence follows.

A landscape asset in a half-canvas slot. The image preserves its native aspect ratio; the lattice pattern frames whatever bands remain. No cropping, ever — authors see the image they dropped in.





Lead with the image, follow with the argument.

A portrait asset in a half-canvas slot — the image fits proportionally, and the lattice pattern shows through on either side. Same rule, different aspect.



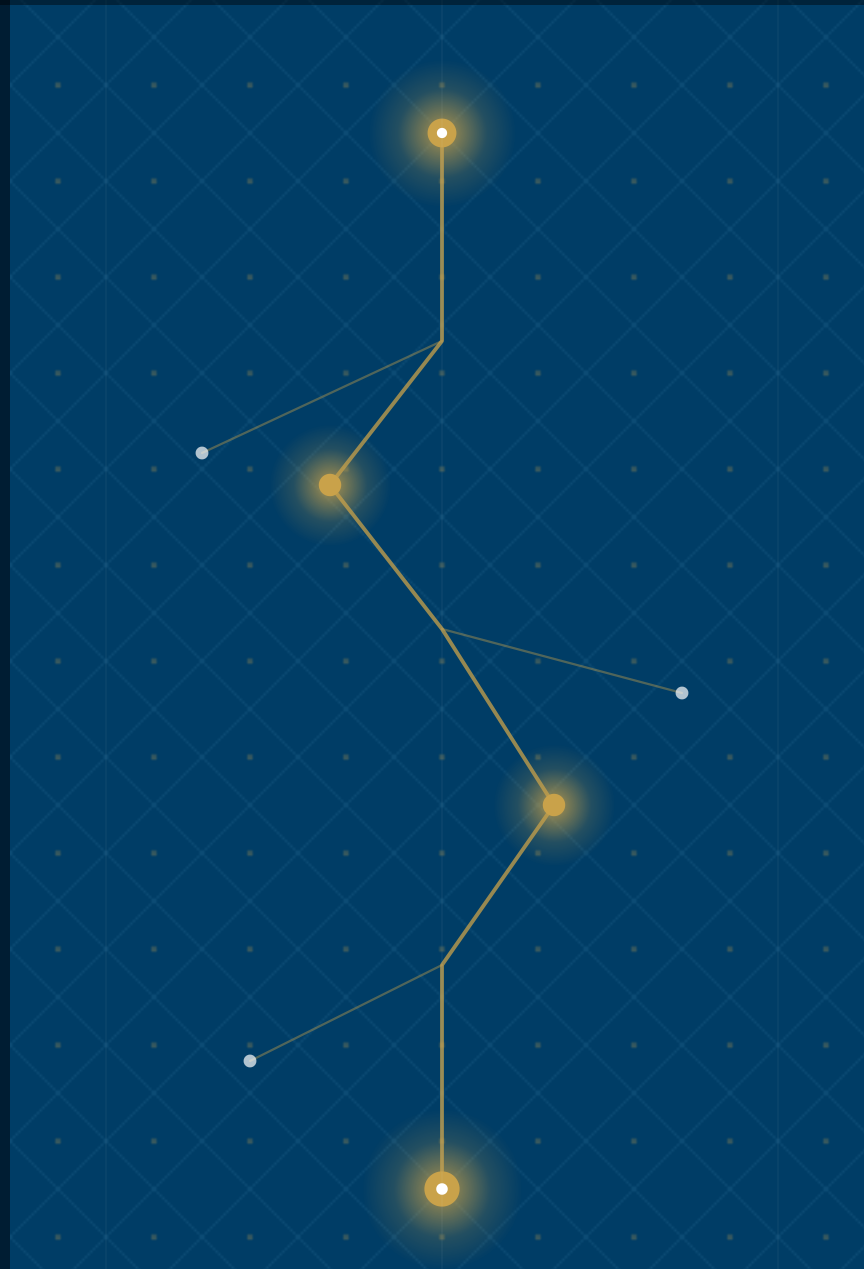
DARK VARIANT · ANY LAYOUT CLASS

The dark modifier works on any layout.

Add *dark* alongside any class — palette remaps automatically

The token system handles dark without per-element overrides.

All colours reference CSS variables — `--bg`, `--text-heading`, `--text-body`, `--border` — that remap when `dark` is added. Cards, headings, body text, and borders all shift automatically. The spectrum bar is suppressed on dark slides.



A tall asset on a wide canvas — the lattice pattern frames the image on the left and right, replacing dead space with brand chrome.

The card stack renders cleanly on dark backgrounds.

Every card uses `--bg-alt` for fill and `--border` for the border — both remap in dark mode.

The accent left border uses `--accent` which is unchanged — the gold reads well against dark.

Body text shifts to `--text-body` which in dark mode is a warm light tone, not pure white.

DARK VARIANT · CARDS STACKED

Two-card layouts work equally well inverted to dark.

The architecture introduces a single key distribution question: what protects the file containing key material, and what is the blast radius if it leaves the host? Every other question in this document depends on the answer.

The pattern here is the same as any page of written argument — claim, then support. The dark palette does not change the information density or the reading rhythm.

The phase modifier renames the prefix word from STEP to PHASE.

PHASE 01

Architecture

The first phase scopes the technical surface — what we build, what we buy, what we defer. Output is an architecture decision record signed by the platform owner.

PHASE 02

Pilot

One internal team, one workload, one quarter. The phase ends when the integration is in production and the on-call rota covers it.

PHASE 03

Rollout

Five teams in two months. The phase ends when no team needs handholding and incident volume is at or below pre-rollout baseline.

MODIFIER · LIST-STEPS MILESTONE LETTERED

Modifiers compose: milestone renames the word, lettered swaps the format.

MILESTONE A

Codebook signing in production

The HSM-anchored signing pipeline runs end-to-end. The first signed codebook installs cleanly on a real client.

MILESTONE B

Multi-tenant DEKs

One codebook can carry distinct DEKs per tenant without per-tenant rebuilds. Crypto-shred is a single HSM op.

MILESTONE C

Per-purpose codebooks

Authoring a codebook scoped to a single business purpose takes minutes, not days. Audit trails distinguish purposes by default.

Vertical stacks the steps as rows; the connector becomes a down-arrow.

STEP 01

Sense

Inputs are signals. Signals are observed, never invented. The first step is to write down what you see, not what you conclude.

STEP 02

Score

A signal becomes data once it carries a number. The score is calibrated against outcomes, not against intuition.

STEP 03

Decide

A decision is a signal plus a deadline. Without the deadline it is an opinion, not a decision. The retrospective closes the loop on the score that earned it.

Chosen flags the right-hand card as the winner.

Vault round-trip

Every detokenize is a network call to a central vault. Latency is a function of distance, not code. p99 60 ms, vault outages cascade.



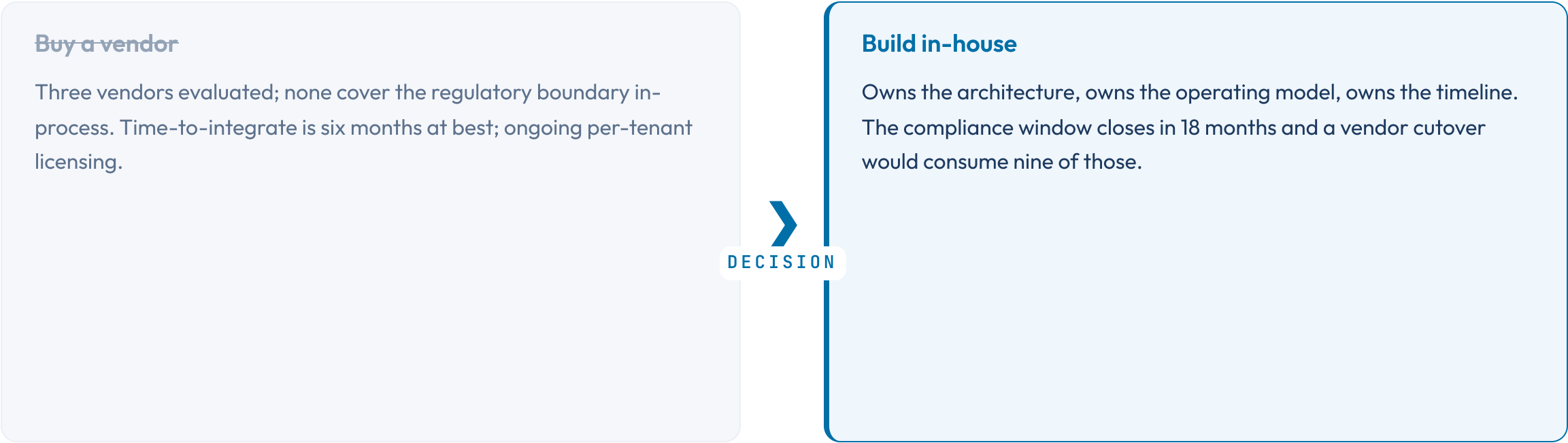
In-process codebook

Detokenize is a local function call against an SDK-resident codebook. p99 8 ms, vault outages do not affect tokenized-record reads.

The right card carries an accent left-edge and accent-tinted background — the same visual contract used by featured cards.

MODIFIER · COMPARE-PROSE DECISION

Decision composes chosen + rejected with a labelled connector.



The left card is struck through to read as the option considered then dropped; the right card carries the chosen visual; the connector is amplified and labelled DECISION.

Vertical stacks the two cards; the arrow connector rotates 90°.

Before — manual rotation

Operators schedule a rotation window, freeze writes on the affected scope, swap codebooks, run a verification pass, lift the freeze. Average outage 18 minutes.



After — version-floor rotation

The signing pipeline emits a new codebook with an incremented version. Clients install the new codebook on next refresh. No write freeze. No coordinated cutover.

Three switches the grid from 2 columns to 3 columns.

Codebook

The signed envelope an SDK installs. Carries policy, wrapped DEK, version, expiry. The codebook is the unit of distribution.

DEK

Data encryption key. Wrapped by a KEK; lives plaintext only inside native SDK memory. Never leaves the host.

KEK

Key encryption key. Lives in the HSM, never exported. The crypto-shred operation on a tenant is a single HSM op against its KEK.

Four switches to 4 columns; pair with compact for visual balance.

Sense

Signals are observed, never invented. Inputs are written down before they are interpreted.

Score

A signal becomes data once it carries a number. Calibration is against outcomes, not intuition.

Decide

A decision is a signal plus a deadline. Without a deadline it is an opinion.

Review

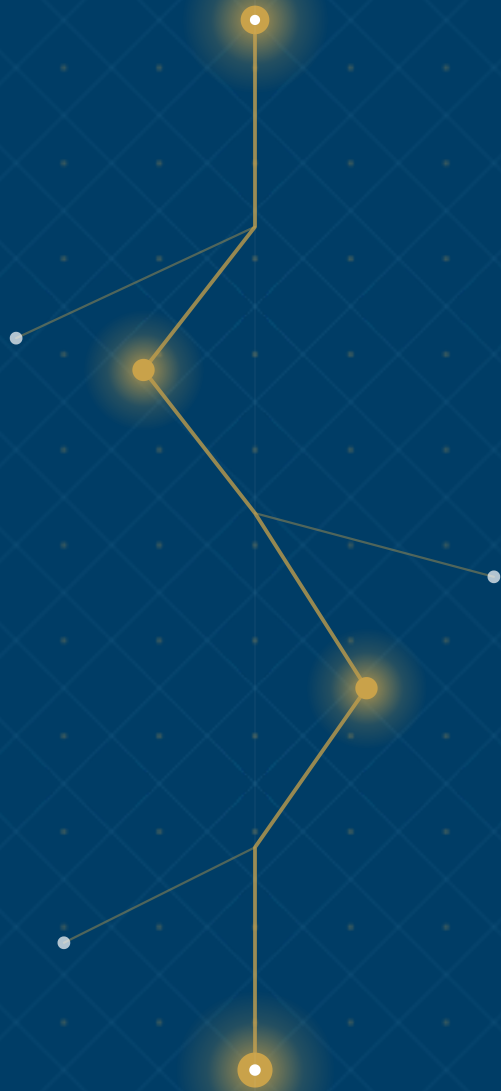
The retrospective closes the loop on the score that earned the decision. The model improves only here.

Horizontal flips cards-stack from a vertical stack to a row.

Claim. The codebook model gets in-process latency with vault-grade key custody. We do not pay round-trip latency on every read.

Evidence. The pilot ran six months across four product teams. p99 detokenize landed at 8 ms; vault outages did not cascade into application outages.

Implication. A vendor cutover is unnecessary. We continue investing in the in-house architecture and ship the operational runbook in the next phase.



MODIFIER · IMAGE MIRROR

Mirror flips the image to the other half — alias of legacy `image left`.

The half-canvas image moves from the right slot to the left slot, and the text padding swaps to match. `mirror` is the cross-cutting orientation flag; `left` remains as a backwards-compatible alias for one release.

Mirror puts the hero card on the right; sub-cards stack on the left.

First supporting card on the left

Useful when the rest of the deck reads left-to-right and the editorial weight needs to land on the right edge as the next slide opens.

Second supporting card below

Identical structure, identical authoring; only the visual side changes.

THE HERO CARD NOW READS FROM THE RIGHT



The featured layout normally leads with the accented hero card on the left and stacks supporting cards on the right. Mirror swaps the columns without touching the markdown contract.

WHAT THIS SECTION COVERS

Mirror moves the dark accent panel to the right. The watermark, eyebrow, and section number all stay anchored to the panel's own box — only the column position flips.

- 1

Confidence
How many independent sources corroborate the signal. Ranges 1–5.
- 2

Recency
Time-decay applied from signal date to scoring date. Half-life is team-configurable.
- 3

Strategic Relevance
Manual score from the signal owner. Ranges 1–5. Requires justification above 4.

SECTION 02 · MIRROR

Section opener with the accent panel on the right.

Mirror composes with chosen — the accented card reads from the left.

The choice

With `mirror`, the chosen card now appears on the left visually. Use this when the surrounding deck reads right-to-left or when the chosen path needs to land first in the audience's scan path.



Considered alternative

Source order keeps this card first, so `chosen` rules continue to target the second card in the markdown. Mirror only flips the rendering; the editorial intent (left = considered, right = chosen) is preserved by reading order.

The below-note still appears under both cards after the hairline rule.

MODIFIER · DIVIDER NUMBERED

Numbered stamps an auto-counter in the top-right corner.

The CSS counter walks the whole deck once and increments on every *divider.numbered* slide. Authors do not number sections by hand — the layout does it.

MODIFIER • SUBTOPIC NUMBERED

Each bookend layout owns its own counter.

The subtopic counter is independent of the divider counter, so a mid-deck subtopic stamps `01` even when the dividers are already at `04`.

CLOSING · NUMBERED

The closing series gets its own auto-stamp too.

Use it for multi-part decks where the closing slide of each part should carry the part number.

How we sort vendors against our two axes.

COVERAGE · COST

High coverage / Low cost

Vendor A — strongest fit on coverage, second-lowest TCO of the four.
Vendor B — narrower coverage but cheapest license tier.

High coverage / High cost

Vendor C — full coverage, premium pricing, niche differentiators we do not need.

Low coverage / Low cost

Vendor D — cheap, but leaves three regulatory boundaries uncovered.

Low coverage / High cost

none — and that is the signal.

We are building, not buying.

DECISION · 2026 Q1

BUILD

Owns the architecture, owns the operating model, owns the timeline.

WHY NOT BUY

Three vendors evaluated; none cover the regulatory boundary in-process.

WHY NOT DELAY

The compliance window closes in 18 months.

Detokenize used to require a vault round-trip.

LATENCY STORY • BEFORE VS AFTER

BEFORE

Every detokenize call: network round-trip to the central vault, average 18 ms, p99 60 ms. Vault outages cascaded into application outages.



AFTER

Detokenize is a local function call. p99 8 ms. Vault outages do not affect tokenized-record reads.

The architecture change is the codebook model — local, signed, time-bound key material — not a vault optimisation.

How we make calls when the spec is silent.

01 We default to the choice that is cheaper to reverse.

02 We name the actor, never the system.

03 We write down the bet on the same slide as the choice.

What ships in each phase, by workstream.

WORKSTREAM	01 PHASE 01	02 PHASE 02	03 PHASE 03
Platform	Codebook signing	Multi-tenant DEKs	Per-purpose codebooks
Operations	Manual rotation	Automated rotation	Crypto-shred
Compliance	Audit trail (HSM)	Centralised log	Examiner pack
SDK	Java	—	Polyglot parity

The first column is sticky workstream label; phase columns carry numbered chrome; empty cells render as a thin dash.

Where we are against quarter targets.

94%

Token-issuance success —
target 99%, +2pp QoQ

8 ms

p99 detokenize —
target 10 ms, -3 ms QoQ

0

Examiner findings —
target 0, flat

3.2×

Detokenize headroom —
target 2×, +0.4× QoQ

What this deck covers, in order.

01 The Design — page 7

02 The Phasing — page 18

03 The Choices — page 26

04 Appendices — page 35

05 Closing — page 64

Who owns each part of the codebook lifecycle.

HSM admin **key custody**

Manages KEK ceremonies and rotation. Never holds plaintext DEKs.

Platform operator **policy**

Owens codebook policy, signing keys, version floors, and revocation playbooks.

Application team **consumption**

Holds time-bound codebooks; tokenizes and detokenizes in-process.

Examiner **oversight**

Reads the HSM audit trail; cannot read plaintext.

SECTION 03 · RECAP

What this section will tell you, in five lines.

01 The codebook model gets in-process latency with vault-grade key custody. → slide 8

02 Rotation is a version-floor increment, not a coordinated cutover. → slide 12

03 Per-tenant KEKs make crypto-shred a single HSM op. → slide 18

04 Phase 1 ships the architecture, Phase 2 ships the operations. → slide 22

05 Five questions stay open until Phase 1 closes them on the record. → slide 27

Compact tightens the spacing scale ~25 %, end-to-end.

What changes

`--sp-xs` through `--sp-2xl` shrink. Card gaps, list gutters, and section padding follow because every layout reads them via `var()`.

What does not change

Type ramp, palette, and chrome reservation (header / footer / pagination) are untouched. Compact is a density flag, not a different layout.

When to reach for it

You have one more card than fits, or your prose runs the section by 1-2 lines, or you want a denser visual rhythm without rewriting copy.

Composition

`compact` composes with `dark`, `accent`, and any layout where density makes sense. It is silently incompatible with `title`, `divider`, and `image-full`.

MODIFIER • LOOSE

Loose is the inverse — more breathing room, same layout machinery.

The spacing scale grows ~25 % rather than shrinks. Sections that already look generous become luxurious; sections that look cramped become balanced. Reach for *loose* when the slide carries a single editorial point and you want the page to feel deliberately quiet — values pages, declarative principles, the closing line of an argument.

The discipline is the same as `compact` from the other side: do not change the type ramp, do not change the chrome, do not change the layout. Only the variables that govern between-element rhythm move.

KEY INSIGHT

Density is not the same as importance. `loose` says: this page deserves room — not because it carries more, but because it carries one thing well.

MODIFIER • ACCENT

Accent replaces the rainbow stripe with a single editorial colour.

The default top border is a spectrum gradient — a system signal that the page is part of a wider deck. The accent modifier swaps that stripe for one solid colour and tints the slide heading. Use it when one slide carries the editorial weight of a section and you want the visual chrome to say so.

It composes with `dark`: on the dark canvas the spectrum top-stripe is suppressed entirely, so `accent.dark` restores a solid accent stripe in its place — preserving the visual signal across both canvases.